Exp.no : 15 **INITIAL EXPLORATION OF DATA**

AIM:

To collect, load, and perform initial exploration of the diabetes dataset.

PYTHON CODE:

```
import pandas as pd
import matplotlib.pyplot as plt
# Load Excel file
df = pd.read_excel("Diabetes.xlsx")
# Display first five rows
print("First Five Rows:")
print(df.head())

# Display dataset shape
print("\nDataset Shape:")
print(df.shape)
# Display column information
print("\nDataset Information:")
df.info()

# Check missing values
print("\nMissing Values:")
print(df.isnull().sum())

# Statistical summary
print("\nStatistical Summary:")
print(df.describe())
# Visualization: Histogram of target values
plt.hist(df["target"], bins=20, edgecolor="black")
plt.title("Distribution of Diabetes Target Values")
plt.xlabel("Target Value")
plt.ylabel("Frequency")
plt.show()
```

OUTPUT:

```
First Five Rows:
        age       sex       bmi  ...        s4        s5        s6
0  0.038076  0.050680  0.061696  ... -0.002592  0.019907 -0.017646
1 -0.001882 -0.044642 -0.051474  ... -0.039493 -0.068332 -0.092204
2  0.085299  0.050680  0.044451  ... -0.002592  0.002861 -0.025930
```

```
3 -0.089063 -0.044642 -0.011595  ...  0.034309  0.022688 -0.009362
4  0.005383 -0.044642 -0.036385  ... -0.002592 -0.031988 -0.046641

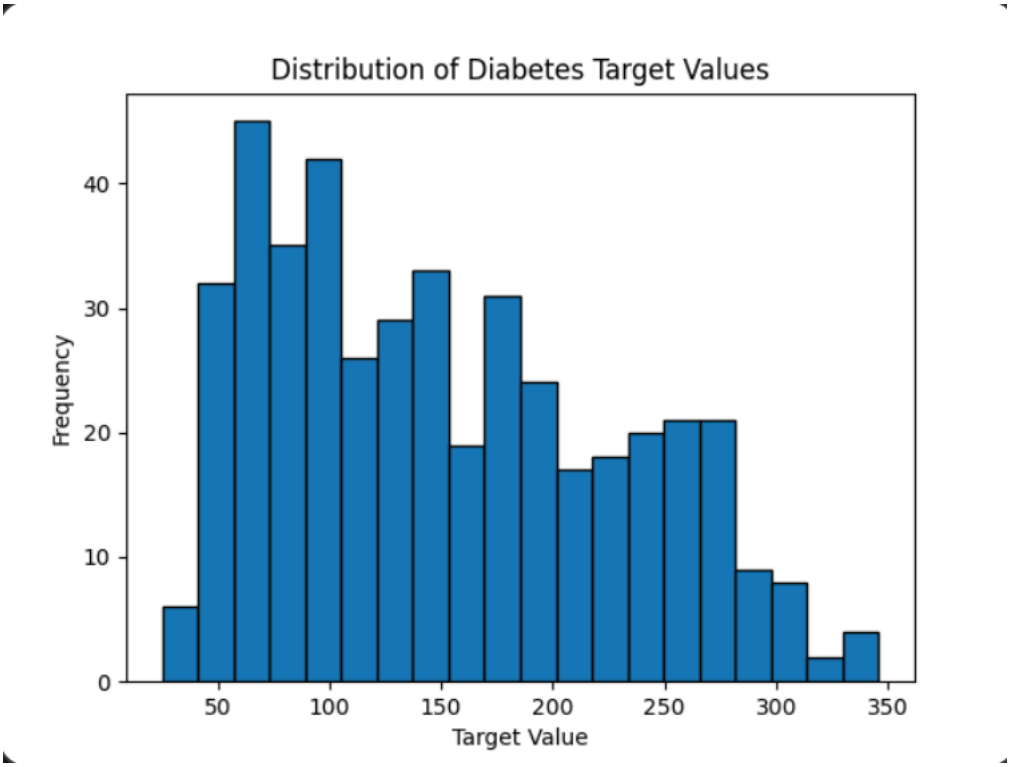
[5 rows x 10 columns]
Dataset Shape:(442, 10)

Column Names:Index(['age', 'sex', 'bmi', 'bp', 's1', 's2', 's3',
's4', 's5', 's6'], dtype='object')

Dataset Information:<class 'pandas.core.frame.DataFrame'>
RangeIndex: 442 entries, 0 to 441
Data columns (total 10 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   age     442 non-null    float64
 1   sex     442 non-null    float64
 2   bmi     442 non-null    float64
 3   bp      442 non-null    float64
 4   s1      442 non-null    float64
 5   s2      442 non-null    float64
 6   s3      442 non-null    float64
 7   s4      442 non-null    float64
 8   s5      442 non-null    float64
 9   s6      442 non-null    float64
dtypes: float64(10)
memory usage: 34.6 KB
None

Missing Values:age    0
sex    0
bmi    0
bp     0
s1     0
s2     0
s3     0
s4     0
s5     0
s6     0
dtype: int64
Statistical Summary:                 age           sex  ...
s5            s6
count  4.420000e+02  4.420000e+02  ...  4.420000e+02  4.420000e+02
mean  -2.511817e-19  1.230790e-17  ...  9.293722e-17  1.130318e-17
std    4.761905e-02  4.761905e-02  ...  4.761905e-02  4.761905e-02
min   -1.072256e-01 -4.464164e-02  ... -1.260971e-01 -1.377672e-01
25%   -3.729927e-02 -4.464164e-02  ... -3.324559e-02 -3.317903e-02
50%    5.383060e-03 -4.464164e-02  ... -1.947171e-03 -1.077698e-03
75%    3.807591e-02  5.068012e-02  ...  3.243232e-02  2.791705e-02
max    1.107267e-01  5.068012e-02  ...  1.335973e-01
1.356118e-01
```

```
[8 rows x 10 columns]
```



RESULT:

The code is executed successfully.